

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

Summary

The tcpdump log shows that the client attempted to resolve the domain www.yummyrecipesforme.com by sending DNS queries over the **UDP** protocol to the DNS server at **203.0.113.2** using **port 53**.

Instead of receiving a DNS response, the client received **ICMP** error messages stating "**udp port 53 unreachable**."

This indicates that the DNS server was unable to receive or process DNS requests because **UDP port 53 was unreachable**.

Protocols involved

- DNS
- UDP
- ICMP
- IP

Details from the log

- Client IP: **192.51.100.15**
- DNS Server IP: **203.0.113.2**
- Destination port: **53**
- DNS query type: **A record**
- Error returned:
 - **ICMP: udp port 53 unreachable**

Interpretation

The DNS service could not be reached because UDP port 53 on the DNS server was unavailable. As a result, the client could not resolve the domain name into

an IP address, preventing access to the website

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The issue was first reported when users were unable to access www.yummyrecipesforme.com and received the message "**destination port unreachable.**"

Scenario, events, and symptoms

Multiple users reported that the website could not be accessed.

The cybersecurity analyst reproduced the problem and captured network traffic using **tcpdump**.

The packet capture showed repeated DNS requests sent over UDP to port 53, followed by ICMP error messages indicating that **UDP port 53 was unreachable.**

Current status

The issue has been identified and reported to the security engineering team for further investigation.

Information discovered

The investigation showed that:

- DNS requests were transmitted correctly.
- The DNS server returned ICMP error messages.
- UDP port 53 was unreachable.
- Because DNS resolution failed, the browser could not obtain the website's IP address.
- Without DNS resolution, the HTTPS connection to the website could not be established.

Next steps

- Verify that the DNS service is running.

- Check whether UDP port 53 is open.
- Review firewall rules blocking DNS traffic.
- Restart the DNS service if necessary.
- Test DNS resolution after corrective actions.

Suspected root cause

The most likely root cause is that the **DNS service on UDP port 53 was unavailable or blocked**, preventing successful DNS queries.